A wardrobe that starts with how you feel.

A local AI outfit recommender using clothes a person already owns.

01 / Problem and hypothesis

Owning many clothes can still leave a person struggling to remember, combine and choose them. Moodrobe explores whether self-reported context and a short, explained selection can reduce this friction and encourage rediscovery. This is a product hypothesis, not a finding from completed interviews.

02 / A complete daily journey

Check in with mood, occasion, temperature and a free-text brief. Browse a photographic sample wardrobe or add personal pieces and local photos. Generate three combinations, inspect the garments, provide preference feedback, save and plan a look, then record an actual wear in the diary.

03 / Designed for inspection

A separate, tab-local sample demo lets a reviewer explore without changing a personal wardrobe. The project brief, model explanation, source model attribution and limitations are one click away. Photographs illustrate pieces; the app does not generate a virtual try-on.

04 / What can be claimed now

Working software, genuine browser-side neural inference and a trainable local preference head. Automated checks cover the main journeys, data persistence, accessibility and real model execution. No user-study participants, accuracy figures or environmental savings have been invented.

Review route: Project brief > Guided sample demo > Inside the AI

The hosted preview remains private. Grant reviewer access before sharing its link.

05 / A hybrid model, with explicit boundaries

A pinned, quantised all-MiniLM-L6-v2 encoder represents the daily brief and garment descriptions as 384-dimensional vectors. ONNX Runtime performs inference in a Web Worker. A structural assembler creates valid owned-item combinations; semantic similarity ranks them, and a regularised logistic preference head learns locally from accepted and rejected suggestions.

06 / What the AI does and does not infer

The pretrained encoder is reused; this is not a foundation model trained from scratch. Rules still constrain outfit completeness and availability. Garment text is the model input, not photographs. Mood is self-reported: camera, voice and RR tools do not provide a validated emotion or health diagnosis. Semantic ranking is not a calibrated probability of a good outfit.

07 / Privacy and engineering tradeoffs

Inference and wardrobe storage run locally, with no paid API key. The first AI run loads roughly 37 MB of local model/runtime assets; performance depends on the device. Personal photos stay in browser storage and deliberate backup files. Backup/restore and deletion are available. Model failures use an explicitly labelled rule-based fallback. Local storage can still be cleared by the browser.

08 / A real usability study is prepared, not completed

The included protocol invites 5-8 consenting adults to set context, run the model, save an outfit, record a wear and inspect limitations. The optional consent flow records actual task events and cumulative elapsed times. Participants decide whether to export anonymous session files. A local dashboard separates sample, mixed and personal wardrobes and starts with zero imported results.

09 / Evidence needed before stronger claims

Collect real observations, report recruitment and limitations, and connect each observed obstacle to a documented change. Ease ratings and small-sample task completion are formative usability evidence. Claims about mood improvement, purchasing or clothing utilisation require a separate appropriately designed longitudinal evaluation.

Reproduce and inspect

Use the guided sample route, compare contrasting briefs and inspect changed outfit rankings. In the source package run npm ci, npm test, npm run test:e2e and npm run build. The release includes the protocol, observation log, a captioned 90-second walkthrough and verification notes.

Model: huggingface.co/Xenova/all-MiniLM-L6-v2 | Sentence-BERT: arxiv.org/abs/1908.10084